GAUGE-ALIGNED ENDPOINTS AND CROSS-DIMENSIONAL TOPOLOGY FOR TEXT EMBEDDING DISTILLATION

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ABSTRACT

Embedding distillation transfers knowledge from a large teacher to a smaller student, yet their representations often differ in dimension and orientation. Existing methods handle this mismatch with trainable projections, intermediate-layer supervision, or auxiliary objectives. We propose GATE-KD, a final-embedding framework centered on a comparable endpoint interface. A fixed principal-component projection selects a teacher subspace at the student dimension, while closed-form orthogonal alignment follows the evolving student across epochs without altering the projected teacher geometry. A complementary topology loss matches connectivity directly between the original teacher and student spaces despite their different dimensions. GATE-KD uses only cached final teacher embeddings and introduces no trainable alignment module. Across three teacher–student configurations and nine downstream benchmarks, GATE-KD obtains the highest aggregate score among the evaluated distilled students, improving over the next-best baseline by 0.38–1.59 points while reducing measured GPU optimization time by 62–74%. Controlled analyses show that alignment matters beyond retained teacher variance and that native-space connectivity complements pointwise endpoint supervision. These results establish a carefully designed endpoint interface as a competitive alternative to more complex distillation pipelines.

1 INTRODUCTION

Text embedding models support retrieval, semantic similarity, clustering, and classification, but the strongest encoders are often too costly for latency- or memory-constrained deployment. Knowledge distillation offers a natural route to compact embedding models by transferring supervision from a frozen teacher to a smaller student (Hinton et al., 2015; Reimers & Gurevych, 2020). Unlike class logits, however, embedding coordinates have no shared semantics across models. Teacher and student representations may differ in dimension, orientation, depth, and even architecture. Existing methods address this mismatch with learned projectors, relational or contrastive objectives, and supervision across intermediate layers (Park et al., 2019; Tian et al., 2020; Xu et al., 2023; Zhang et al., 2024a; Truong et al., 2025; Dao et al., 2026). Intermediate alignment can provide rich supervision, but it presupposes useful correspondences between teacher and student depths—an assumption that becomes fragile as their architectures diverge—and often requires additional mappings or repeated access to teacher hidden states (Ko et al., 2023; Yu et al., 2025). More fundamentally, this machinery can obscure whether the difficulty lies in coordinating the representation hierarchy or in failing to define a comparable final representation in the first place. This motivates our hypothesis that the endpoint interface is an under-examined bottleneck in cross-dimensional distillation.

We test this hypothesis by separating teacher knowledge into two channels. The *extrinsic interface* describes how teacher representations are expressed in coordinates that the student can match pointwise, whereas *intrinsic structure* comprises properties of the representation cloud that do not depend on its coordinates or ambient dimension. This decomposition leads to **GATE-KD**, Gauge Alignment and Topology for Embedding Distillation. GATE-KD fits a fixed principal-component subspace to cached teacher endpoints, selecting the information that can pass through the student-width bottleneck. Within that subspace, an orthogonal Procrustes update aligns the target coordinates to the current student and is refitted after each epoch on a fixed calibration subset. Because each update is orthogonal, it changes only the targets' coordinate gauge, leaving their pairwise geometry unchanged. In parallel, GATE-KD matches the degree-zero persistent-homology signatures of student batches and the original, unprojected teacher batches. This second term transfers multiscale connectivity directly between native spaces of different dimensions. The resulting objective combines pointwise endpoint correspondence with coordinate-invariant structure while supervising only the final student embedding. It trains no projector, uses no intermediate-layer objective, and requires no online teacher forward pass during ordinary training.

Experiments across three teacher–student configurations and nine downstream benchmarks support this endpoint-centered view. Under endpoint-only supervision, adding the Procrustes-aligned gauge to the same PCA subspace raises the average score from 69.35 to 74.28, showing that retained teacher variance alone does not determine whether a target is easy to learn. Native-space H_0 supervision then improves the endpoint-only result from 74.28 to 74.79, indicating that intrinsic connectivity provides a complementary refinement. With both signals active, refitting the gauge across epochs adds 0.31 points over retaining the alignment fitted at initialization. Overall, GATE-KD achieves the highest aggregate score among the evaluated distilled students in all three configurations, exceeding TALAS by 0.38, 0.48, and 1.59 points while using only 26–38% of its measured GPU optimization time. These results show that, in the settings we study, strong embedding transfer does not require coordinating the intermediate representation hierarchy when the endpoint interface and native-space structure are specified explicitly.

We make three contributions.

- We formulate cross-dimensional embedding distillation as an intrinsic–extrinsic decomposition, separating the coordinate interface required for pointwise supervision from structure that can be transferred directly across native spaces.
- We introduce GATE-KD, a terminal-only method that combines a fixed PCA subspace, epoch-wise closed-form gauge alignment, and native-space H_0 persistence matching using only cached final teacher embeddings and no trainable alignment module.
- Through controlled interface and signal decompositions, we show that coordinate comparability and topology provide complementary supervision; across three teacher–student configurations, this simple formulation improves aggregate performance while substantially reducing measured optimization cost relative to a strong hierarchy-based baseline.

2 Related Work

Early text-embedding distillation directly regressed student sentence vectors onto teacher outputs, showing that unlabeled text can provide dense supervision without task annotations (Reimers & Gurevych, 2020). Subsequent methods have enriched this signal through contrastive learning, specialized training recipes, and teacher-compatible output spaces (Xu et al., 2023; Zhang et al., 2024a; Li et al., 2025; Vujanic & Rückstieß, 2026). EmbedDistill combines embedding matching with relative geometric supervision for information retrieval (Kim et al., 2023). Other approaches draw on the representation hierarchy. EMO transfers token-level relations and aligns hidden states across tokenizers, while TALAS anchors upper student layers to the teacher endpoint and propagates supervision through lower layers by self-distillation (Truong et al., 2025; Dao et al., 2026). Intermediate supervision can provide additional signals, but it also introduces choices about layer correspondence and often requires teacher hidden states or auxiliary mappings. Prior analyses have identified overfitting in intermediate-layer distillation and found that its layer-selection strategy may have limited influence (Ko et al., 2023; Yu et al., 2025). GATE-KD studies the complementary regime in which the teacher is accessed only through cached final embeddings and every training signal is applied to the student endpoint.

Dimension mismatch in feature distillation is commonly handled by mapping one representation space into the other. HPD learns projection layers that produce compact sentence representations (Zhao et al., 2022), and broader studies show that a projector's parameterization and normalization can substantially change the student's optimization dynamics (Miles & Mikolajczyk, 2024; Chen et al., 2023). Orthogonally constrained projections preserve intra-batch similarity, while TCS uses PCA coordinates extracted from cached teacher features (Miles et al., 2024; Zhou et al., 2025). More generally, representation-similarity work treats orthogonal transformations as natural equivalences because they preserve inner products and distances, with Procrustes providing the corresponding alignment (Kornblith et al., 2019; Williams et al., 2021; Maystre et al., 2025). Recent work formalizes teacher supervision over representation equivalence classes (Han, 2026). GATE-KD instantiates this perspective for cross-dimensional text embeddings through a terminal-only interface combined with native-space H_0 transfer. Most closely related to our geometric motivation, Bhattarai et al. (2025) show that unconstrained projected matching can attain zero feature loss without preserving the teacher's Gram geometry. Under their assumptions, zero Procrustes distance is instead equivalent to zero Gram discrepancy. GATE-KD instead uses Procrustes to estimate a coordinate gauge on a fixed calibration set. The gauge is held constant within each epoch and updated in closed form as the student evolves. PCA selects the retained teacher subspace, while the orthogonal update changes its orientation without altering the projected teacher geometry.

Coordinate-free objectives offer another route to transferring representation structure. Relational distillation matches pairwise distances or angles, similarity-preserving distillation matches Gram structure, and contrastive distillation transfers relations through positive and negative examples (Park et al., 2019; Tung & Mori, 2019; Tian et al., 2020). Persistent homology extends this view from pairwise relations to multiscale properties of a point cloud. TopKD transfers global feature topology through approximate persistence images, while topological-guidance distillation combines image-derived persistence features from multiple teachers (Kim et al., 2024; Jeon et al., 2024). GATE-KD uses topology neither as a surrogate for all representation geometry nor as a replacement for sample-wise supervision. It directly matches the sorted degree-zero persistence death times of student batches and unprojected teacher batches, which requires neither shared coordinates nor equal ambient dimension. The resulting decomposition places coordinate-dependent correspondence behind an explicit endpoint interface while allowing coordinate-invariant connectivity to transfer directly across native spaces.

3 Method

Let $f_T : \mathcal{X} \to \mathbb{R}^{d_T}$ be a frozen teacher and $f_S : \mathcal{X} \to \mathbb{R}^{d_S}$ a student with $d_T > d_S$. We run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache its normalized final sentence embeddings

$$\mathbf{t}_i = \mathrm{norm}(f_T(x_i)) \in \mathbb{S}^{d_T-1}, \tag{1}$$

where $\mathrm{norm}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$. The corresponding student embedding is $\mathbf{z}_i = \mathrm{norm}(\mathrm{Pool}(h_i^{(L)})) \in \mathbb{S}^{d_S-1}$. GATE-KD transfers teacher knowledge through two terminal channels. An extrinsic interface expresses each teacher endpoint in coordinates that the student can match pointwise, while an intrinsic objective transfers structure that does not require shared coordinates or equal ambient dimension.

3.1 Extrinsic Endpoint Interface

Subspace selection. Let $\mu = N^{-1} \sum_i \mathbf{t}_i$. We fit the leading d_S principal directions of the centered cache and collect them as $P_{\mathrm{PCA}} \in \mathbb{R}^{d_T \times d_S}$, with $P_{\mathrm{PCA}}^\top P_{\mathrm{PCA}} = I_{d_S}$. Centering is used to estimate the principal directions; in the evaluated recipe, the cached vectors are not centered when the projection is applied. The normalized coordinates before gauge alignment are

$$\mathbf{y}_i = \mathrm{norm}(\mathbf{t}_i P_{\mathrm{PCA}}) \in \mathbb{S}^{d_S-1}. \tag{2}$$

The PCA subspace is fitted once on the full teacher cache and remains fixed throughout training.

Gauge alignment. The basis used to express the retained subspace affects a pointwise loss even though it does not affect the geometry within that subspace. We resolve this coordinate ambiguity with orthogonal Procrustes alignment (Schönemann, 1966; Bhattarai et al., 2025). A calibration subset $\mathcal{C}$ is selected before training and reused unchanged. Let $Y_{\mathcal{C}}$ contain its projected teacher coordinates and let $Z_{\mathcal{C}}^{(e)}$ contain the current student embeddings at the start of epoch e. We compute

$$R^{(e)} = \underset{R \in O(d_S)}{\arg\min} \left\| Y_{\mathcal{C}} R - Z_{\mathcal{C}}^{(e)} \right\|_F^2 = UV^\top, \quad U\Sigma V^\top = \mathrm{svd}\left(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)} \right). \tag{3}$$

The initialized student determines $R^{(0)}$. After each non-final epoch, we re-encode the same calibration sentences and refit the gauge for the following epoch. The target used within epoch e is therefore

$$\boldsymbol{\tau}_i^{(e)} = \mathbf{y}_i R^{(e)}. \tag{4}$$

Each target is fixed within an epoch. The refit is a closed-form coordinate update rather than a learned projector.

Relation to the endpoint objective. Because the rows of $Y_{\mathcal{C}}$ and $Z_{\mathcal{C}}^{(e)}$ have unit norm, their cosine endpoint objective satisfies

$$\frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \left(1 - \left\langle \mathbf{z}_i^{(e)}, \mathbf{y}_i R \right\rangle\right) = \frac{1}{2|\mathcal{C}|} \left\| Z_{\mathcal{C}}^{(e)} - Y_{\mathcal{C}} R \right\|_F^2 . \tag{5}$$

Equation 3 therefore exactly minimizes the calibration endpoint loss over R at each refit boundary with the student fixed.

Gauge invariance. For every orthogonal R,

$$(YR)(YR)^\top = YY^\top. \tag{6}$$

Consequently, refitting the gauge leaves all pairwise cosines, chord distances, and persistent-homology signatures of the projected teacher targets unchanged. PCA determines which teacher subspace survives the width bottleneck, while Procrustes determines how that fixed geometry is presented to the evolving student.

Corollary (Monotone gauge refit)**.** *With the student fixed, every gauge refit weakly decreases the calibration endpoint loss, while leaving the topology term unchanged.*

Proof. Equation 3 minimizes the endpoint loss over all $R \in O(d_S)$, so its value cannot exceed that obtained with the previous gauge. The topology term depends only on the fixed native teacher and student embeddings and is independent of R. $\square$

3.2 Intrinsic Topology Transfer

The endpoint interface can reproduce only the geometry retained after projection. Although P_{PCA} has orthonormal columns, $P_{\mathrm{PCA}} P_{\mathrm{PCA}}^\top$ is a rank-d_S projector, so the cosine Gram matrix of the normalized coordinates in Equation 2 need not equal that of the original teacher embeddings. We therefore transfer an additional structural signal directly from the teacher's native space.

This loss of geometry is unavoidable whenever the normalized teacher cache has rank above the student width.

Proposition 1 (Irreducible Gram distortion)**.** *Let $T \in \mathbb{R}^{N \times d_T}$ collect the normalized teacher cache embeddings, with singular values $\sigma_1(T) \geq \cdots \geq \sigma_r(T) > 0$. For every $Z \in \mathbb{R}^{N \times d_S}$ collecting student embeddings of the same N sentences,*

$$\left\| TT^\top - ZZ^\top \right\|_F^2 \geq \sum_{j > d_S} \sigma_j(T)^4. \tag{7}$$

Thus exact preservation of the teacher cosine Gram matrix is impossible when $\operatorname{rank}(T) > d_S$.

Proof. The eigenvalues of $TT^\top$ are $\sigma_j(T)^2$, whereas $\operatorname{rank}(ZZ^\top) \leq d_S$. The Eckart–Young theorem therefore lower-bounds the squared error of any such rank-d_S approximation by the squared tail eigenvalues (Eckart & Young, 1936). $\square$

For a minibatch of B normalized vectors, we use the chord distance

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v} \rangle}. \tag{8}$$

The $B - 1$ finite death times of degree-zero Vietoris–Rips persistence are the edge weights of the point cloud's minimum spanning tree (Hofer et al., 2020). Let $\delta_{(1)}^S \leq \cdots \leq \delta_{(B-1)}^S$ and $\delta_{(1)}^T \leq$

$\cdots \leq \delta^T_{(B-1)}$ denote the sorted death times of the student embeddings $\{\mathbf{z}_i\}$ and the original teacher embeddings $\{\mathbf{t}_i\}$. We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left(\delta^S_{(j)} - \delta^T_{(j)} \right)^2 . \tag{9}$$

For a precise interpretation, define the empirical distributions of finite connectivity scales

$$\nu_S = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{\delta^S_{(j)}}, \qquad \nu_T = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{\delta^T_{(j)}}, \tag{10}$$

where δ_a denotes a Dirac mass at a.

Proposition 2 (Transport interpretation and stability)**.** *The H_0 loss equals the squared one-dimensional Wasserstein distance between the empirical death-scale measures,*

$$\mathcal{L}_{H_0} = W_2^2(\nu_S, \nu_T). \tag{11}$$

Moreover, if the two distance matrices on the corresponding batch satisfy

$$\epsilon = \max_{i,j} \left| d(\mathbf{z}_i, \mathbf{z}_j) - d(\mathbf{t}_i, \mathbf{t}_j) \right|, \tag{12}$$

then $\mathcal{L}_{H_0} \leq \epsilon^2$.

Proof. Optimal transport on the real line pairs equally weighted samples in sorted order, which gives Equation 11 (Peyré & Cuturi, 2019). Under a uniform edge perturbation of at most ϵ, every threshold graph at scale a in either metric is contained in the other at scale $a + \epsilon$. Kruskal's characterization of the ordered minimum-spanning-tree edges then gives $|\delta^S_{(j)} - \delta^T_{(j)}| \leq \epsilon$ for every j, and averaging their squared differences proves the bound. This is the degree-zero finite-space case of Vietoris–Rips persistence stability (Chazal et al., 2014). ☐

The teacher diagram is computed from $\mathbf{t}_i \in \mathbb{R}^{d_T}$ rather than $\mathbf{y}_i$ or $\boldsymbol{\tau}_i^{(e)}$. The loss therefore needs neither a projection nor a shared orientation. It is piecewise differentiable through the selected student edge lengths (Hofer et al., 2019). The stability result is one-way. Equation 9 matches the multiset of H_0 death times but does not impose correspondence between individual minimum-spanning-tree edges, recover the full metric geometry, or preserve higher-dimensional topological features.

3.3 Objective and Alternating Optimization

The endpoint loss for a training minibatch is

$$\mathcal{L}^{(e)}_{\text{end}} = \frac{1}{B} \sum_{i=1}^{B} \left(1 - \left\langle \mathbf{z}_i, \boldsymbol{\tau}_i^{(e)} \right\rangle \right). \tag{13}$$

GATE-KD optimizes the two-term objective

$$\mathcal{L}^{(e)} = \mathcal{L}^{(e)}_{\text{end}} + \lambda_{\text{topo}} \mathcal{L}_{H_0}. \tag{14}$$

Here, $\mathcal{L}_{H_0}$ is evaluated on the same minibatch using the student embeddings and the corresponding unprojected teacher embeddings.

Optimization alternates between stochastic updates of the student parameters and exact updates of the coordinate gauge. During epoch e, $R^{(e)}$ is held fixed while the student is optimized with Equation 14. After epoch e, unless it is the final epoch, the updated student re-encodes $\mathcal{C}$ and Equation 3 produces $R^{(e+1)}$ for the next epoch. The H_0 term is unaffected by these gauge updates because it is computed from the native teacher embeddings.

Teacher embeddings and the PCA subspace are computed once before student optimization. Ordinary training steps use the cached teacher embeddings and require no online teacher forward pass. Each gauge update adds one student pass over the calibration set and an SVD of a $d_S \times d_S$ matrix.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Teacher–student configurations. We adopt the three teacher–student pairs evaluated by TALAS (Dao et al., 2026). They comprise Qwen3-Embedding-4B → BERT-base, BGE-M3 → MiniLMv2-L6-H768, and Qwen3-Embedding-0.6B → MiniLMv2-L6-H384 (Devlin et al., 2019; Wang et al., 2021; Chen et al., 2024; Zhang et al., 2025). These settings span students from 22M to 109M parameters and embedding bottlenecks of $2560 \to 768$, $1024 \to 768$, and $1024 \to 384$, respectively. The Qwen teachers use last-token pooling, whereas BGE-M3 uses CLS pooling.

Distillation corpus. We construct the distillation corpus by sampling approximately 5K raw texts from each of the Emotion, WiC, and STS-B training sets. No labels or task annotations are used during distillation. We regard these three benchmarks as in-domain (IOD) and the remaining six evaluation benchmarks as out-of-domain (OOD). Every method uses the same corpus.

Baselines. We compare GATE-KD with methods covering complementary transfer mechanisms. RKD matches inter-example relations (Park et al., 2019), while Stella uses a multi-stage embedding-distillation recipe (Zhang et al., 2024a). CDM and DSKD address cross-tokenizer alignment and dual-space transfer, respectively (Chen et al., 2025; Zhang et al., 2024b). EMO combines token-level relations with hidden-state alignment (Truong et al., 2025), and TALAS combines teacher anchoring with hierarchy-based self-distillation (Dao et al., 2026). Method-specific training hyperparameters are reported in Appendix A.

Evaluation protocol. We evaluate frozen student embeddings on nine benchmarks from three task families. For classification on Banking77, Emotion, and Tweet, we fit a logistic-regression probe on each benchmark's training split and report macro-F1. For pair classification on MRPC, SciTail, and WiC, we map cosine similarity to $[0, 1]$ and report average precision. For semantic textual similarity on SICK, STS12, and STS-B, we report Spearman correlation with the gold scores. All metrics are multiplied by 100. Avg. is the unweighted mean of the nine primary scores, while Avg. IOD and Avg. OOD average the three in-domain and six out-of-domain benchmarks, respectively. We report means and sample standard deviations over three seeds.

4.2 MAIN RESULTS

Table 1 compares GATE-KD with six distillation baselines across nine benchmarks and three teacher–student configurations. GATE-KD achieves the highest overall average in all three configurations, improving over the next-best baseline by 0.38, 0.48, and 1.59 points in configurations (a), (b), and (c), respectively. The advantage is largest for the 22M-parameter student and therefore does not diminish in the most capacity-constrained configuration.

The task-level results show that this advantage is broad but not uniform. GATE-KD ranks first on 18 of the 27 individual benchmark comparisons, with its most consistent results on classification and semantic textual similarity. Other methods remain stronger on some pair-classification benchmarks, including MRPC and WiC. In the largest capacity-gap configuration, GATE-KD essentially matches TALAS in-domain, with 72.04 versus 72.19 Avg. IOD, while improving Avg. OOD from 79.85 to 80.49. The aggregate gain in this configuration is therefore driven by OOD performance rather than higher scores on the datasets supplying the distillation text.

4.3 OPTIMIZATION EFFICIENCY

Table 2 reports observed optimization cost under the same execution protocol on NVIDIA H200 GPUs. Step time and throughput exclude the first ten warm-up steps. GPU training time sums the measured optimization-step durations and excludes teacher-cache construction, tokenization, and final evaluation.

GATE-KD uses 26%, 31%, and 38% of TALAS's measured GPU optimization time in configurations (a), (b), and (c), respectively, while also attaining higher overall scores. The improvement is

Table 1: Main results across nine benchmarks. Distilled-student scores are mean ± sample standard deviation over three seeds. Best and second-best results are shown in bold and underlined, respectively.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Averages		
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	IOD	OOD	All
(a) Qwen3-Embedding-4B → BERT-base (109M)												
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	74.25	84.48	81.07
Student base	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	42.43	60.08	54.20
RKD	89.76± 0.11	70.90± 0.31	59.98± 0.05	81.94± 0.26	75.59± 0.13	69.02± 0.18	72.93± 0.07	67.84± 0.18	73.72± 0.08	67.57± 0.05	76.49± 0.07	73.52± 0.06
Stella	88.48± 0.30	68.94± 0.17	53.69± 0.50	83.95± 0.10	75.39± 0.54	**71.93**± 0.09	69.87± 0.64	64.39± 0.21	70.86± 0.36	65.49± 0.17	75.17± 0.13	71.95± 0.06
CDM	88.13± 0.23	69.03± 0.20	52.83± 0.50	83.00± 0.06	77.10± 0.27	71.84± 0.09	67.89± 0.37	61.06± 0.37	69.19± 0.42	64.62± 0.28	74.37± 0.12	71.12± 0.18
DSKD	87.97± 0.16	68.90± 0.28	53.07± 0.31	83.03± 0.15	76.98± 0.59	71.86± 0.08	67.77± 0.66	61.11± 0.26	69.55± 0.12	64.83± 0.12	74.29± 0.14	71.14± 0.10
EMO	86.89± 0.07	68.99± 0.11	52.02± 1.09	79.74± 0.12	76.37± 0.35	69.62± 0.16	67.89± 0.09	56.46± 0.47	64.54± 0.50	62.06± 0.49	72.72± 0.15	69.17± 0.24
TALAS	90.53± 0.09	73.77± 0.21	68.47± 0.34	**85.38**± 0.05	**81.83**± 0.21	69.59± 0.14	76.79± 0.06	70.79± 0.19	**78.50**± 0.08	**72.19**± 0.07	79.85± 0.07	77.29± 0.04
GATE-KD	**90.60**± 0.21	**73.86**± 0.10	**69.48**± 0.22	85.31± 0.11	80.23± 0.13	68.19± 0.12	**77.80**± 0.14	**75.15**± 0.07	78.44± 0.16	72.04± 0.08	**80.49**± 0.03	**77.67**± 0.03
(b) BGE-M3 → MiniLMv2-H768 (66M)												
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	71.78	83.81	79.80
Student base	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	44.62	62.24	56.36
RKD	89.14± 0.09	73.29± 0.09	62.23± 0.27	84.35± 0.06	78.74± 0.11	65.02± 0.10	73.20± 0.19	65.02± 0.29	71.97± 0.38	66.41± 0.07	77.29± 0.06	73.66± 0.04
Stella	86.82± 0.28	70.06± 0.12	56.61± 0.74	83.02± 0.12	74.63± 0.12	**69.72**± 0.05	69.08± 0.20	63.14± 0.28	67.10± 0.10	64.47± 0.27	74.46± 0.05	71.13± 0.11
CDM	87.82± 0.19	69.99± 0.09	58.32± 0.51	83.44± 0.07	75.56± 0.40	69.70± 0.02	65.93± 0.16	60.84± 0.34	65.31± 0.15	64.45± 0.13	73.93± 0.06	70.77± 0.02
DSKD	87.70± 0.16	70.35± 0.32	59.71± 0.61	82.99± 0.18	75.58± 0.12	69.59± 0.04	65.36± 0.37	61.21± 0.31	65.02± 0.18	64.77± 0.27	73.87± 0.10	70.83± 0.11
EMO	87.54± 0.18	71.03± 0.14	58.52± 0.30	82.46± 0.18	74.73± 0.58	69.20± 0.22	64.71± 0.55	56.89± 0.27	61.25± 0.03	62.99± 0.19	72.89± 0.10	69.59± 0.12
TALAS	89.83± 0.09	**74.70**± 0.09	61.65± 0.23	**86.62**± 0.05	80.85± 0.22	66.72± 0.07	75.40± 0.10	67.08± 0.31	77.08± 0.01	68.48± 0.07	79.08± 0.08	75.55± 0.06
GATE-KD	**90.17**± 0.05	73.74± 0.24	**64.83**± 0.72	86.42± 0.09	**80.97**± 0.08	65.47± 0.09	**76.77**± 0.03	**68.27**± 0.04	**77.63**± 0.12	**69.31**± 0.21	**79.39**± 0.06	**76.03**± 0.11
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)												
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	72.41	82.50	79.14
Student base	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	41.52	58.16	52.62
RKD	86.76± 0.07	69.87± 0.21	56.31± 0.35	81.70± 0.03	76.03± 0.20	68.56± 0.22	69.21± 0.18	64.26± 0.19	70.25± 0.21	65.04± 0.20	74.64± 0.09	71.44± 0.10
Stella	84.04± 0.04	68.73± 0.44	52.41± 0.28	81.50± 0.12	72.26± 0.32	**68.65**± 0.16	64.64± 0.19	60.59± 0.12	64.27± 0.20	61.78± 0.13	71.96± 0.13	68.57± 0.10
CDM	84.74± 0.21	69.52± 0.20	53.76± 0.95	81.15± 0.14	72.86± 0.30	68.29± 0.11	63.98± 0.36	60.43± 0.22	63.52± 0.24	61.86± 0.43	72.11± 0.19	68.69± 0.27
DSKD	84.76± 0.10	69.56± 0.20	53.14± 0.29	81.15± 0.01	72.81± 0.09	68.33± 0.04	63.78± 0.06	60.24± 0.19	63.53± 0.18	61.67± 0.03	72.05± 0.06	68.59± 0.04
EMO	83.54± 0.31	69.31± 0.06	54.48± 0.27	80.87± 0.13	72.00± 0.16	67.89± 0.09	62.19± 0.16	57.04± 0.17	61.24± 0.23	61.20± 0.09	70.82± 0.07	67.62± 0.02
TALAS	86.39± 0.13	71.81± 0.11	58.40± 0.30	**84.25**± 0.03	79.29± 0.11	66.82± 0.02	70.42± 0.10	68.08± 0.22	73.34± 0.10	66.19± 0.11	76.71± 0.04	73.20± 0.06
GATE-KD	**87.65**± 0.06	**72.32**± 0.24	**63.04**± 0.28	84.20± 0.06	**81.60**± 0.05	68.07± 0.12	**72.48**± 0.08	**68.96**± 0.18	**74.76**± 0.14	**68.62**± 0.21	**77.87**± 0.02	**74.79**± 0.06

Table 2: Training efficiency on NVIDIA H200 GPUs.

Method	Avg. ↑	ms/step ↓	samples/s ↑	GPU train min ↓	peak GiB ↓
(a) Qwen3-Embedding-4B → BERT-base (109M)					
RKD	73.52 ± 0.06	119.7 ± 18.3	1081.0 ± 179.8	1.20 ± 0.18	1.72 ± 0.00
Stella	71.95 ± 0.06	130.7 ± 8.1	975.9 ± 58.8	1.27 ± 0.08	1.78 ± 0.01
CDM	71.12 ± 0.18	340.4 ± 13.6	374.2 ± 15.3	3.38 ± 0.12	9.31 ± 0.01
DSKD	71.14 ± 0.10	290.9 ± 9.0	437.7 ± 13.3	2.88 ± 0.09	9.45 ± 0.00
EMO	69.17 ± 0.24	721.1 ± 12.9	176.5 ± 3.1	7.02 ± 0.12	9.30 ± 0.00
TALAS	77.29 ± 0.04	329.0 ± 12.5	387.1 ± 15.0	3.23 ± 0.11	**1.43** ± 0.00
GATE-KD	**77.67** ± 0.03	**83.4** ± 3.8	**1527.2** ± 70.8	**0.84** ± 0.04	1.72 ± 0.00
(b) BGE-M3 → MiniLMv2-H768 (66M)					
RKD	73.66 ± 0.04	77.5 ± 8.8	1656.4 ± 201.2	0.79 ± 0.09	1.07 ± 0.00
Stella	71.13 ± 0.11	152.8 ± 30.8	859.1 ± 195.6	1.49 ± 0.30	1.09 ± 0.00
CDM	70.77 ± 0.02	127.5 ± 13.8	1006.4 ± 115.8	1.30 ± 0.12	2.15 ± 0.01
DSKD	70.83 ± 0.11	115.0 ± 21.9	1137.2 ± 241.0	1.14 ± 0.20	2.22 ± 0.01
EMO	69.59 ± 0.12	394.6 ± 33.4	324.0 ± 28.3	3.87 ± 0.35	2.15 ± 0.01
TALAS	75.55 ± 0.06	184.4 ± 4.6	690.2 ± 17.3	1.82 ± 0.03	**0.85** ± 0.00
GATE-KD	**76.03** ± 0.11	**56.5** ± 7.7	**2281.6** ± 310.0	**0.57** ± 0.07	1.07 ± 0.00
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)					
RKD	71.44 ± 0.10	64.8 ± 13.7	2012.3 ± 488.6	0.68 ± 0.14	0.41 ± 0.00
Stella	68.57 ± 0.10	150.7 ± 15.3	850.0 ± 81.5	1.47 ± 0.13	0.42 ± 0.00
CDM	68.69 ± 0.27	161.8 ± 21.6	796.3 ± 113.1	1.65 ± 0.20	1.53 ± 0.00
DSKD	68.59 ± 0.04	175.4 ± 8.2	726.4 ± 33.5	1.78 ± 0.06	1.56 ± 0.00
EMO	67.62 ± 0.02	499.3 ± 1.4	254.8 ± 0.7	4.90 ± 0.01	1.53 ± 0.00
TALAS	73.20 ± 0.06	169.8 ± 15.4	753.7 ± 71.4	1.69 ± 0.14	**0.34** ± 0.00
GATE-KD	**74.79** ± 0.06	**62.9** ± 3.1	**2025.2** ± 98.3	**0.64** ± 0.04	0.41 ± 0.00

primarily in throughput rather than peak memory allocation, for which TALAS remains lower across the three configurations.

5 ANALYSIS

5.1 COORDINATE COMPARABILITY AT THE BOTTLENECK

Table 3 isolates the endpoint interface in configuration (c) by holding the endpoint-only objective and training protocol fixed. Within the same PCA subspace, the unaligned coordinates reach 69.35 Avg., a random orthogonal orientation reaches 70.37, and the fitted Procrustes gauge reaches 74.28. Procrustes alignment therefore improves the unaligned PCA interface by 4.93 points without changing either the retained subspace or its pairwise geometry. The fitted gauge also exceeds the random orientation by 3.91 points, indicating that its benefit comes from alignment with the student rather than from rotating the target coordinates alone.

The fixed PCA–Procrustes interface also outperforms learned maps in both directions, which reach 61.46 for teacher-to-student mapping and 69.33 for student-to-teacher mapping. This result is consistent with avoiding a mapping that is updated jointly with the student by gradient descent. Retained energy alone does not explain downstream performance. The random subspace retains 0.381 of the teacher energy, compared with 0.928 for PCA, yet exceeds unaligned PCA by 1.57 points. Together, these controls show that a high-energy subspace is not an effective pointwise target unless its coordinates are made comparable to the student. The random controls use a single map draw, so they do not establish a general ranking between PCA and randomly selected subspaces.

Table 3: Endpoint-interface ablation in configuration (c). All variants optimize $\mathcal{L}_{\mathrm{end}}$ only. Scores are mean $\pm$ sample standard deviation over three seeds. Random controls use one map draw shared across the training seeds.

Target map	Gauge	Retained energy ↑	Avg. IOD ↑	Avg. OOD ↑	Avg. ↑
Learned $T \to S$	—	N/A	54.34$\pm$ 0.68	65.02$\pm$ 0.25	61.46$\pm$ 0.39
Learned $S \to T$	—	N/A	62.54$\pm$ 0.36	72.72$\pm$ 0.31	69.33$\pm$ 0.32
PCA	None	0.928	62.49$\pm$ 0.19	72.78$\pm$ 0.07	69.35$\pm$ 0.04
PCA	Random orthogonal	0.928	63.68$\pm$ 0.10	73.72$\pm$ 0.09	70.37$\pm$ 0.08
Random orthogonal	None	0.381	64.44$\pm$ 0.14	74.16$\pm$ 0.04	70.92$\pm$ 0.07
PCA	**Procrustes**	**0.928**	**67.76**$\pm$ 0.05	**77.54**$\pm$ 0.07	**74.28**$\pm$ 0.03

5.2 COMPLEMENTARY ENDPOINT AND TOPOLOGY SIGNALS

Table 4 separates coordinate-dependent endpoint supervision from coordinate-invariant topology transfer. The endpoint-only objective provides most of the downstream improvement, raising Avg. from 52.62 for the base student to 74.28. In contrast, H_0-only supervision reaches 64.22 while reducing the topology residual from 0.3595 to 0.1240. The low residual but substantially lower downstream score shows that matching connectivity alone does not determine semantic correspondence. Combining the two signals improves the endpoint-only result by 0.51 points to 74.79 and lowers its topology residual from 0.2122 to 0.1281. Endpoint correspondence and topological structure therefore provide complementary supervision.

The remaining comparisons evaluate two refinements within the combined objective. Computing H_0 from the original teacher space improves Avg. from 74.57 to 74.79 relative to projected teacher targets and produces a lower topology residual, 0.1281 versus 0.1355. Refitting the gauge after each epoch improves Avg. from 74.48 to 74.79 over retaining the alignment fitted at initialization. Both gains are concentrated more strongly in IOD performance and are modest relative to the contribution of the endpoint interface. These results support native-space topology and epoch-wise refitting as refinements of the complete objective rather than its primary source of improvement.

Table 4: Decomposition of endpoint supervision, topology transfer, and gauge refitting in configuration (c). Scores are mean $\pm$ sample standard deviation over three seeds. The H_0 residual is measured in the original teacher space.

Configuration	H_0 source	Gauge refit	H_0 residual ↓	Avg. IOD ↑	Avg. OOD ↑	Avg. ↑
Student base	—	—	0.3595$\pm$ 0.0000	41.52$\pm$ 0.00	58.16$\pm$ 0.00	52.62$\pm$ 0.00
H_0 only	Original	—	**0.1240**$\pm$ 0.0012	58.00$\pm$ 1.01	67.33$\pm$ 0.62	64.22$\pm$ 0.73
Endpoint only	—	On	0.2122$\pm$ 0.0004	67.76$\pm$ 0.05	77.54$\pm$ 0.07	74.28$\pm$ 0.03
Endpoint $+H_0$	Projected	On	0.1355$\pm$ 0.0003	68.27$\pm$ 0.12	77.72$\pm$ 0.00	74.57$\pm$ 0.04
Endpoint $+H_0$	Original	Off	0.1309$\pm$ 0.0001	68.03$\pm$ 0.10	77.71$\pm$ 0.03	74.48$\pm$ 0.01
Endpoint $+H_0$ (**Ours**)	Original	On	0.1281$\pm$ 0.0003	**68.62**$\pm$ 0.21	**77.87**$\pm$ 0.02	**74.79**$\pm$ 0.06

5.3 Sensitivity to Training Choices

Figure 1 examines sensitivity to the topology weight, training batch size, and gauge-calibration size in configuration (c). Performance remains stable across moderate topology weights, with $\lambda_{\text{topo}} = 0.5$ attaining the highest average and settings from 0.5 to 1.0 differing by at most 0.24 points. Reducing the gauge-calibration set from the full corpus to 2,048 examples changes Avg. by only 0.28 points, indicating that the alignment can be estimated from a relatively small sample. Smaller batches obtain higher scores, but this sweep fixes the number of epochs rather than optimizer updates. The batch comparison is therefore not compute matched and does not establish that smaller batches are intrinsically better. Complete numerical results are reported in Appendix D.

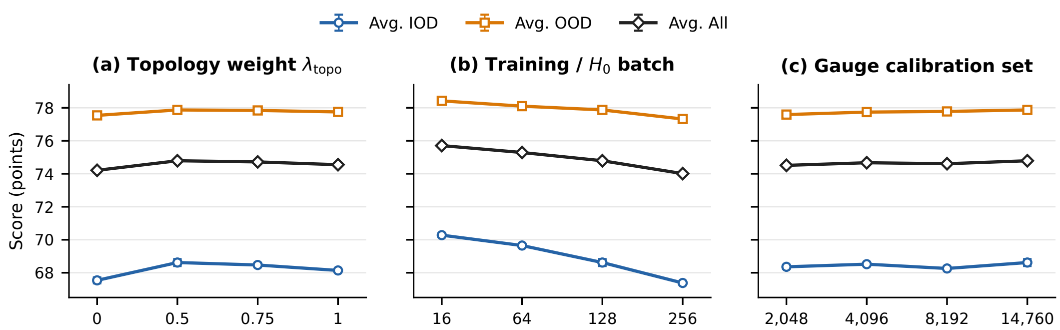

Figure 1: Sensitivity to topology weight, training batch size, and gauge-calibration size in configuration (c). Points and error bars show mean $\pm$ sample standard deviation over three seeds. The batch sweep fixes the number of epochs and is not compute matched.

6 Conclusion

Cross-dimensional embedding distillation is not only a compression problem but also requires a well-defined interface between teacher and student coordinates. GATE-KD addresses this issue by combining a fixed PCA subspace with closed-form gauge alignment and native-space H_0 matching, using only final embeddings. Across three teacher–student configurations, this terminal-only formulation achieves the highest aggregate scores among the evaluated distilled students while substantially reducing measured optimization time relative to a strong hierarchy-based baseline. Controlled analyses show that coordinate alignment matters beyond retained variance and that topology complements pointwise endpoint supervision without requiring a shared ambient space. These results support treating coordinate-dependent correspondence and coordinate-invariant structure as separate design problems in embedding distillation. The current sentence-level objective does not fully preserve the teacher's zero-shot retrieval capability, leaving broader retrieval transfer for future work.

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A Training Details

All runs use a maximum sequence length of 256 and seeds 42, 43, and 44. For GATE-KD, $\lambda_{\text{topo}} = 0.5$, the H_0 cloud is the current training batch, and the gauge is refitted after each non-final epoch. Gauge calibration uses the full distillation corpus. Table 5 lists the method-specific training hyperparameters.

Table 5: Training hyperparameters for the main experiments.

Method	Epochs	Batch size	Learning rate
RKD	5	128	7×10^{-5}
Stella	$2 + 3$	128	5×10^{-5}
CDM	5	128	2×10^{-5}
DSKD	5	128	2×10^{-5}
EMO	5	128	1×10^{-5}
TALAS	5	128	2×10^{-5}
GATE-KD	5	128	7×10^{-5}

B Results at a Larger Distillation Scale

We repeat the complete method sweep on approximately 100K unlabeled texts drawn from the training splits of all nine sentence-level benchmarks. No labels or task annotations are used. Relative to the main comparison, this setting changes both corpus size and dataset coverage, so it evaluates robustness to broader distillation data rather than isolating data scale alone.

Table 6: Results at a larger distillation scale.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Avg.
(a) Qwen3-Embedding-4B → BERT-base (109M)										
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	81.07
Student base	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	54.20
RKD	92.23± 0.15	71.96± 0.18	63.79± 0.40	85.00± 0.04	80.27± 0.38	66.31± 0.10	79.32± 0.02	71.72± 0.16	77.70± 0.20	76.48± 0.04
Stella	90.02± 0.15	71.90± 0.22	60.14± 0.45	85.96± 0.05	78.82± 0.27	69.93± 0.13	76.53± 0.23	69.79± 0.12	77.22± 0.16	75.59± 0.10
CDM	89.16± 0.04	69.63± 0.23	53.74± 0.13	85.57± 0.08	71.22± 0.19	**70.04**± 0.05	71.26± 0.29	66.97± 0.21	73.69± 0.38	72.36± 0.13
DSKD	88.87± 0.07	69.05± 0.09	54.02± 0.56	85.90± 0.05	70.02± 0.31	69.63± 0.21	71.07± 0.10	66.52± 0.17	73.52± 0.01	72.07± 0.03
EMO	84.96± 0.84	68.93± 0.65	54.80± 1.52	85.05± 0.04	65.75± 0.22	68.80± 0.13	62.00± 2.71	61.75± 1.85	69.54± 0.37	69.07± 0.89
TALAS	93.12± 0.13	**74.67**± 0.38	71.42± 0.26	**86.30**± 0.01	87.06± 0.12	68.09± 0.02	80.29± 0.03	76.34± 0.01	82.26± 0.05	79.95± 0.08
GATE-KD	**93.47**± 0.12	74.18± 0.25	**72.50**± 0.47	85.45± 0.08	**87.24**± 0.12	66.57± 0.08	**81.85**± 0.06	**76.40**± 0.07	**82.52**± 0.08	**80.02**± 0.03
(b) BGE-M3 → MiniLMv2-H768 (66M)										
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	79.80
Student base	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	56.36
RKD	91.68± 0.06	74.57± 0.40	62.45± 0.75	85.81± 0.08	79.81± 0.17	63.57± 0.13	76.91± 0.05	69.54± 0.11	75.56± 0.09	75.54± 0.10
Stella	88.72± 0.41	73.43± 0.23	61.63± 0.29	86.39± 0.08	74.13± 0.31	66.98± 0.27	73.32± 0.07	68.30± 0.12	72.66± 0.14	73.95± 0.04
CDM	87.64± 0.23	70.82± 0.14	56.37± 0.57	85.67± 0.08	67.69± 0.23	66.89± 0.20	70.15± 0.48	66.68± 0.26	69.49± 0.15	71.27± 0.10
DSKD	86.69± 0.38	69.76± 0.22	56.98± 0.39	84.92± 0.07	66.75± 0.17	**67.23**± 0.10	68.26± 0.45	66.28± 0.02	68.31± 0.35	70.58± 0.10
EMO	87.17± 0.59	71.07± 0.22	59.32± 0.02	84.25± 0.41	67.89± 0.92	66.09± 0.18	69.04± 0.38	62.26± 1.77	66.47± 0.40	70.40± 0.35
TALAS	92.44± 0.10	75.58± 0.15	65.13± 0.21	**87.09**± 0.09	85.39± 0.14	64.39± 0.12	78.14± 0.06	72.56± 0.13	**81.03**± 0.05	77.97± 0.03
GATE-KD	**92.56**± 0.04	**76.30**± 0.31	**65.62**± 0.29	86.58± 0.05	**87.20**± 0.06	63.43± 0.02	**79.23**± 0.03	**72.69**± 0.05	80.60± 0.13	**78.24**± 0.05
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)										
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	79.14
Student base	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	52.62
RKD	89.74± 0.04	72.34± 0.22	57.99± 0.52	84.13± 0.12	75.20± 0.14	**67.49**± 0.06	74.48± 0.03	68.12± 0.15	72.55± 0.14	73.56± 0.09
Stella	85.74± 0.22	70.25± 0.07	53.48± 0.28	84.43± 0.04	71.23± 0.05	66.51± 0.14	70.96± 0.28	63.02± 0.04	67.42± 0.24	70.34± 0.02
CDM	85.18± 0.08	69.14± 0.07	52.30± 0.26	84.02± 0.04	68.24± 0.23	66.57± 0.29	69.15± 0.13	63.05± 0.06	66.55± 0.10	69.36± 0.08
DSKD	85.04± 0.14	69.21± 0.13	52.85± 0.36	84.34± 0.10	67.33± 0.24	65.98± 0.12	69.01± 0.13	63.46± 0.21	66.42± 0.09	69.29± 0.09
EMO	85.08± 0.17	68.11± 0.10	53.96± 0.60	84.38± 0.16	65.48± 0.16	65.23± 0.17	68.61± 0.07	60.00± 0.82	65.38± 0.38	68.47± 0.15
TALAS	89.79± 0.03	73.87± 0.17	61.75± 0.61	**85.99**± 0.03	82.07± 0.09	65.79± 0.04	75.67± 0.05	**73.91**± 0.03	78.31± 0.04	76.35± 0.08
GATE-KD	**91.68**± 0.09	**74.43**± 0.21	**66.88**± 0.36	85.00± 0.06	**83.46**± 0.02	67.44± 0.04	**77.94**± 0.06	72.17± 0.02	**78.36**± 0.07	**77.48**± 0.04

Table 6 shows that GATE-KD attains the highest mean Avg. in all three teacher–student configurations. It exceeds TALAS by 0.07, 0.27, and 1.13 points in configurations (a), (b), and (c), respectively. The margin is narrow for the two larger students and remains largest for the compact student.

C Zero-Shot Retrieval

We evaluate the models distilled in the main comparison on ArguAna, SciFact, SciDocs, NFCorpus, and FiQA-2018 using exhaustive cosine ranking. Evaluation uses no instruction prefix or task-specific retrieval training and reports nDCG@10. Retrieval scores are not included in the sentence-level averages of Table 1. Teacher and student-base scores are deterministic, while distilled-student scores are mean ± sample standard deviation over three seeds.

Table 7: Zero-shot retrieval on five BEIR benchmarks after distillation, measured by nDCG@10.

Method	ArguAna	SciFact	SciDocs	NFCorpus	FiQA-2018	Avg.
(a) Qwen3-Embedding-4B → BERT-base (109M)						
Teacher	74.66	73.75	25.69	36.23	50.15	52.10
Student base	12.65	0.89	0.38	1.33	0.05	3.06
RKD	39.68 ± 0.25	9.74 ± 0.64	1.19 ± 0.04	4.30 ± 0.30	2.82 ± 0.09	11.55 ± 0.18
Stella	39.12 ± 0.17	14.29 ± 0.17	<u>3.79</u> ± 0.26	6.21 ± 0.14	5.57 ± 0.04	13.79 ± 0.03
CDM	33.62 ± 0.27	15.11 ± 0.20	3.28 ± 0.14	6.26 ± 0.09	5.08 ± 0.22	12.67 ± 0.04
DSKD	33.70 ± 0.39	15.75 ± 0.49	3.35 ± 0.11	6.21 ± 0.17	5.34 ± 0.24	12.87 ± 0.27
EMO	21.90 ± 0.34	8.84 ± 0.28	1.87 ± 0.11	4.87 ± 0.16	2.04 ± 0.06	7.91 ± 0.07
TALAS	**48.95** ± 0.17	**19.94** ± 0.09	**5.70** ± 0.12	**7.94** ± 0.08	**8.15** ± 0.02	**18.14** ± 0.07
GATE-KD	<u>45.20</u> ± 0.32	<u>19.25</u> ± 0.66	2.81 ± 0.09	<u>6.79</u> ± 0.22	<u>6.56</u> ± 0.21	<u>16.12</u> ± 0.26
(b) BGE-M3 → MiniLMv2-H768 (66M)						
Teacher	54.01	64.16	16.37	31.80	41.12	41.49
Student base	9.32	1.15	0.44	1.22	0.08	2.44
RKD	21.27 ± 0.23	<u>7.84</u> ± 0.12	<u>1.96</u> ± 0.05	<u>4.31</u> ± 0.14	2.26 ± 0.03	7.53 ± 0.09
Stella	18.46 ± 0.37	3.11 ± 0.27	1.25 ± 0.06	2.29 ± 0.15	1.66 ± 0.08	5.35 ± 0.16
CDM	15.83 ± 0.19	3.77 ± 0.38	0.88 ± 0.02	2.88 ± 0.16	1.41 ± 0.21	4.95 ± 0.15
DSKD	16.25 ± 0.15	4.05 ± 0.10	0.88 ± 0.03	2.96 ± 0.11	1.59 ± 0.07	5.15 ± 0.07
EMO	13.10 ± 0.27	2.83 ± 0.17	0.72 ± 0.04	2.25 ± 0.07	0.85 ± 0.09	3.95 ± 0.08
TALAS	**30.14** ± 0.32	6.55 ± 0.34	1.90 ± 0.08	2.82 ± 0.18	<u>2.83</u> ± 0.09	<u>8.85</u> ± 0.17
GATE-KD	<u>25.30</u> ± 0.02	**11.22** ± 0.27	**3.31** ± 0.02	**4.93** ± 0.06	**4.83** ± 0.01	**9.92** ± 0.05
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)						
Teacher	68.11	68.26	15.72	29.73	38.68	44.10
Student base	9.85	0.66	0.32	1.28	0.05	2.43
RKD	30.12 ± 0.79	<u>5.57</u> ± 0.76	<u>1.77</u> ± 0.12	<u>3.34</u> ± 0.07	<u>1.88</u> ± 0.06	8.54 ± 0.33
Stella	24.49 ± 0.45	2.65 ± 0.09	1.09 ± 0.07	2.62 ± 0.07	1.01 ± 0.11	6.37 ± 0.09
CDM	21.09 ± 0.30	1.66 ± 0.17	1.13 ± 0.03	2.57 ± 0.05	0.89 ± 0.02	5.47 ± 0.06
DSKD	21.56 ± 0.38	1.77 ± 0.20	1.18 ± 0.06	2.63 ± 0.03	0.89 ± 0.07	5.61 ± 0.13
EMO	17.18 ± 0.14	1.41 ± 0.09	0.92 ± 0.02	1.96 ± 0.06	0.70 ± 0.03	4.44 ± 0.05
TALAS	<u>35.21</u> ± 0.52	4.17 ± 0.09	1.44 ± 0.04	3.12 ± 0.10	1.87 ± 0.07	<u>9.16</u> ± 0.14
GATE-KD	**36.18** ± 0.09	**10.26** ± 0.21	**3.44** ± 0.06	**4.30** ± 0.06	**3.57** ± 0.03	**11.55** ± 0.06

GATE-KD achieves the highest retrieval average in configurations (b) and (c). In configuration (a), it ranks second to TALAS, with 16.12 versus 18.14 Avg. All distilled students remain substantially below their teachers, indicating that the sentence-level distillation objectives do not fully preserve the teachers' zero-shot retrieval capability.

D DETAILED SENSITIVITY RESULTS

Table 8 reports the numerical results underlying Figure 1. Settings not varied within each block remain at the main recipe. The H_0 cloud is always the current training batch. The batch sweep fixes the number of epochs and is therefore not compute matched.

Table 8: Numerical sensitivity results for configuration (c).

Factor	Value	Avg. IOD ↑	Avg. OOD ↑	Avg. ↑
λ_{topo}	0	67.54± 0.18	77.54± 0.07	74.21± 0.03
	0.5	**68.62**± 0.21	**77.87**± 0.02	**74.79**± 0.06
	0.75	68.47± 0.13	77.84± 0.04	74.72± 0.04
	1	68.14± 0.06	77.75± 0.03	74.55± 0.03
Training/H_0 batch	16	**70.28**± 0.09	**78.42**± 0.09	**75.71**± 0.03
	64	69.65± 0.09	78.10± 0.08	75.29± 0.05
	128	68.62± 0.21	77.87± 0.02	74.79± 0.06
	256	67.38± 0.03	77.32± 0.07	74.01± 0.05
Gauge calibration	2,048	68.36± 0.11	77.59± 0.04	74.51± 0.06
	4,096	68.52± 0.08	77.74± 0.05	74.67± 0.02
	8,192	68.26± 0.08	77.78± 0.03	74.61± 0.02
	14,760	**68.62**± 0.21	**77.87**± 0.02	**74.79**± 0.06